Graph Sketching: Hyperbolic Functions

Questions 1–12

Question 1. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

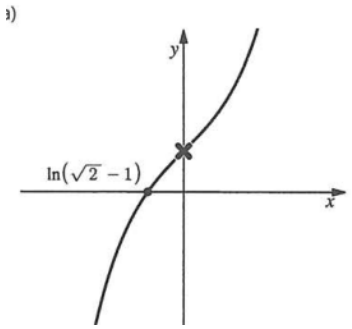
a) $y = \sinh x$
b) $y = \cosh x$
c) $y = \tanh x$

Question 2. Sketch the following pairs of graphs on the same axes. Label each graph clearly and show where they intersect the coordinate axes:

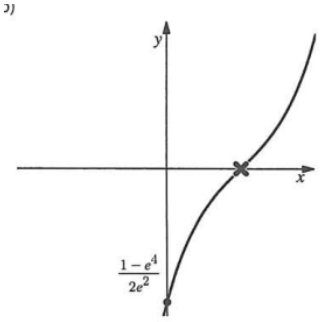
a) $y = \sinh x$ and $y = 2\sinh x$
b) $y = \sinh x$ and $y = \sinh(x + 3)$
c) $y = \sinh x$ and $y = -\sinh x$
d) $y = \cosh x$ and $y = \cosh x + 1$
e) $y = \cosh x$ and $y = \cosh 2x$
f) $y = \cosh x$ and $y = -\cosh x$

Question 3. By considering the point of inflection (identified on the graphs by a ×), find the equation of each of the following graphs in the form $y = \sinh(x - a)$ or $y = \sinh x + b$, where a and b are integers:

a)

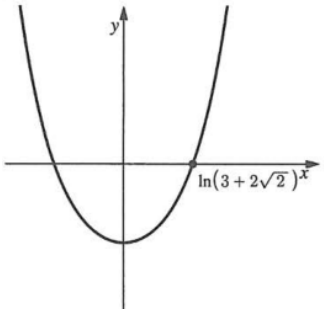


b)

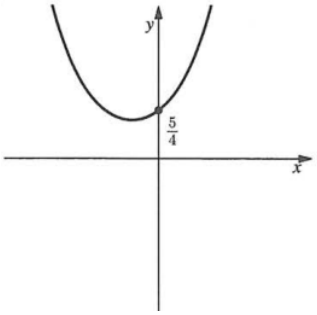


Question 4. Find the equation of each of the following graphs in the form $y = \cosh(x - a)$ or $y = \cosh x + b$, where a and b are integers:

a)



b)



Question 5. State the equations of the asymptotes of the following graphs:

a) $y = \tanh(x - 2) + 5$

b) $y = \tanh(x + \frac{5}{2}) - \frac{3}{2}$

Question 6. Sketch each of the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \tanh x + 1$

b) $y = \tanh(x - 1)$

c) $y = \tanh x - 2$

d) $y = -\tanh x$

Question 7. Sketch each of the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = 3\tanh x$

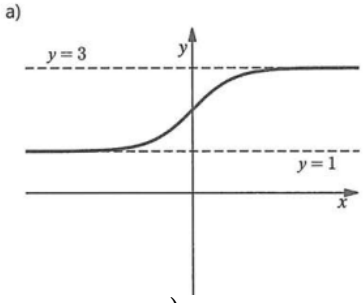
b) $y = \frac{1}{2}\tanh x$

Question 8. Find the equation of each of the following graphs in the form $y = \tanh(x - a)$ or $y = \tanh x + b$, where a and b are integers:

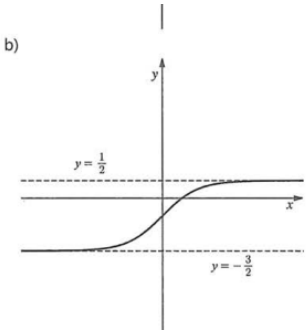
a)

$y = \tanh x + b$, where a and b are integers:

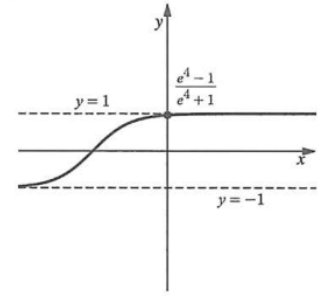
a)



b)



c)



Question 9. It is given that $k > 1$. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

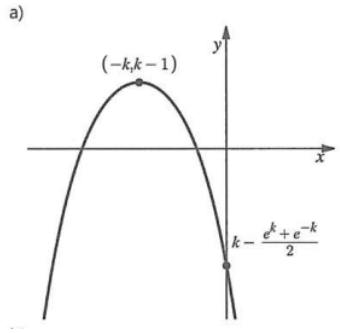
a) $y = \sinh(x - k) + k$
b) $y = \cosh(x + k) - k$
c) $y = \tanh(x - k) - k$

Question 10. Sketch each of the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

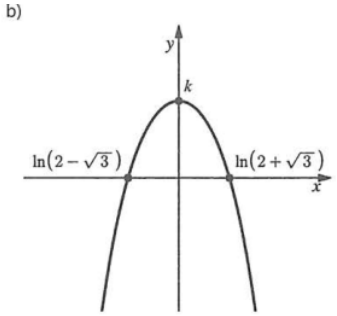
a) $y = 1 - \sinh x$
b) $y = -\frac{1}{2}\cosh x$
c) $y = 2\tanh x + 1$
d) $y = 2\sinh x - 1$
e) $y = 3\cosh x - 5$
f) $y = 2 - \frac{1}{2}\tanh x$

Question 11. Both of the following hyperbolic curves are transformations of $y = \cosh x$. No horizontal stretches have been performed. For constant k where $k > 1$, find a suitable equation for each:

a)



b)



Question 12. It is given that $1 < k < 2$. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \sinh(kx - 2k)$
b) $y = 3k - \cosh(x + k)$
c) $y = 4k - k\tanh(x - k)$